

Advancing Dense Retrieval Architectures in Retrieval-Augmented Generation

Elena Vance, Marcus Thorne, Alistair Finch — Department of AI Systems, Horizon University

Abstract

Retrieval-Augmented Generation (RAG) has emerged as the premier paradigm to ground Large Language Models (LLMs) on non-parametric knowledge. However, traditional sparse lexical matching frequently suffers from vocabulary mismatch when encountering nuanced academic queries. In this work, we investigate dense bi-encoder architectures paired with Sentence-Transformer embeddings and FAISS vector indexing. We evaluate multi-stage dense retrieval over 150,000 academic passages and demonstrate that cosine similarity in normalized 384-dimensional dense semantic space achieves a 14.2% improvement in Mean Reciprocal Rank (MRR@10) over BM25.

1. Introduction

Parametric LLM memory is prone to knowledge cutoff and catastrophic hallucination. RAG mitigates this limitation by retrieving authoritative external passages dynamically. The core challenge lies in the semantic retrieval phase: how to transform heterogeneous user queries into dense numerical vectors that accurately align with dense document representations in vector databases.

2. Dense Embedding Methodology

Our proposed framework utilizes a dual-encoder transformer architecture (e.g., all-MiniLM-L6-v2) trained with contrastive infoNCE loss. Given a query q and document chunk d , semantic similarity is computed as $S(q, d) = \cos(\|E(q)\|, \|E(d)\|)$. To enable sub-millisecond retrieval across millions of chunks, we employ FAISS with Inverted File Indexing (IVF) and Hierarchical Navigable Small World (HNSW) graphs.

3. Experimental Evaluation and Results

We conducted comparative retrieval benchmarks across four benchmark datasets (MS-MARCO, SciFact, HotpotQA, and BioASQ). Our dense FAISS pipeline attained Recall@5 of 88.4% and Recall@10 of 94.1%. The query latency averaged 4.2 milliseconds per search on a single CPU core, proving that lightweight dense representations offer an optimal balance of throughput and accuracy.

4. Advantages and Strengths

1. Superior semantic capture: Dense embeddings capture conceptual synonyms and paraphrase variations where keyword search fails completely.
2. Scalable indexing: FAISS vector indices allow near-linear scaling to millions of embeddings with low memory footprint.
3. Modular integration: Embeddings can be refreshed without retraining the underlying language generation model.

5. Limitations and Future Scope

Despite strong semantic recall, pure dense retrieval occasionally fails on exact alphanumeric identifiers, version codes, and chemical formulas. Future research should explore hybrid sparse-dense ensembles and dynamic context compression mechanisms.

6. Conclusion

Dense retrieval using Sentence Transformers and FAISS provides an efficient, robust, and mathematically grounded foundation for multi-document RAG pipelines.